

Juan Carlos Verduzco

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West Lafayette, IN

PROFESSIONAL SUMMARY

Materials Engineering Ph.D. with expertise in data science, machine learning, and materials informatics. Research work focused on materials development for solid-state batteries. Equipped with a background in mechatronics engineering, paired with robust electrical engineering experience.

EDUCATION

Doctor of Philosophy – Materials Engineering

2018 – 2023

Purdue University, West Lafayette, IN Computational Interdisciplinary Graduate Program

Bachelor of Science – Mechatronics Engineering

2013 – 2017

Monterrey Institute of Technology and Higher Education (ITESM), Zapopan, MEX

Study Abroad program on Artificial Intelligence at Universitat Politècnica de Catalunya (Spain)

PROFESSIONAL EXPERIENCE

Lead Software Engineer

2026 – Present

Purdue University, West Lafayette, IN

- Directed the design and implementation of infrastructure and data analytics strategies on nanoHUB, overseeing user behavior, content, and scientific data pipelines to advance NCN's strategic university, industry, and government partnerships.
- Led the adoption of FAIR data principles and state-of-the-art LLM applications across nanoHUB's cyberinfrastructure, driving data interoperability and enabling cutting-edge research, education, and workforce development initiatives.
- Supervised a team of data professionals while developing external funding proposals, expanding NCN's capacity for data infrastructure investment and ensuring continued platform growth across the global science and engineering community.

Senior Data Science Engineer

2023 – 2026

Purdue University, West Lafayette, IN

- Spearheaded the development of infrastructure and analytics tools on nanoHUB, focusing on user behavior, content, and scientific data to support the strategic objectives of both SCALE and MEST projects.
- Employed data visualization to extract and communicate insights from complex datasets, enabling informed decision-making.
- Leveraged advanced software engineering techniques for the testing and implementation of data science, machine learning, and computational materials science tools in nanoHUB. These tools have been used by 3,825 unique users and account for over 29,000 individual simulation runs.

Graduate Research Assistant

2018 – 2023

Purdue University, West Lafayette, IN

- Implemented data-driven machine learning pipelines to accelerate materials discovery for organic solar cells, battery materials, and high temperature metal alloys.
- Characterized and tested electrochemical properties of composite polymer electrolyte films, identifying key opportunities for enhanced safety and flexibility.
- Conducted molecular dynamics simulations using parallel programming in HPC supercomputer clusters and analyzed mechanisms for enhancing ionic conductivity in battery materials.

Test Engineering Technician III

2018 – 2018

Jabil Circuit Mexico, Zapopan, MEX

- Directed a diagnostics testing team of 8 engineers specialized in electrical components and manufacturing process FMEA.
- Built and managed documentation and tracking systems to improve manufacturing line processes.

RELEVANT SKILLS AND LANGUAGES

Programming Languages

- Proficient: Python, LaTeX
- Familiar: C++, SQL, MATLAB
- Tools: SLURM, Git/SVN

Materials Characterization

- Techniques: XRD, DSC, EIS, SEM, Raman Spectroscopy
- Computational Packages: VASP, LAMMPS

Machine Learning and Data Science

- Frameworks: TensorFlow/Keras, PyTorch
- Libraries: Pandas, Scikit-learn, Plotly

Software

- Design: SolidWorks, CATIA
- Others: LabVIEW, Microsoft Office Suite

Languages

- English (Fluent)
- Spanish (Fluent)
- German (Beginner)